

Research Proposal: Learning to Predict Strong Moves for Solving a 3×3 Rubik's Cube

Project Goal

The goal of this project is to train a neural network that can predict strong or optimal next moves for solving a 3×3 Rubik's Cube. We want the model to learn from cube states at different distances from the solved state. The training will start from simple states, such as states that are only one move away from being solved, and then gradually increase the difficulty up to states that may require around 20 moves to solve.

Motivation

The Rubik's Cube is an interesting problem in Artificial Intelligence because it has a very large state space, but it also has a clear goal state and a set of legal actions. A known result about the 3×3 Rubik's Cube is that any solvable cube state can be solved in at most 20 moves, assuming the standard move metric where a half turn is counted as one move. This makes the cube a good environment for studying search, planning, and learning-based decision making.

Traditional solving methods usually rely on hand designed algorithms, search methods, or mathematical properties of the cube. In this project, we want to explore a learning based approach. Instead of explicitly programming all the solving rules, we want the neural network to learn patterns from examples of cube states and good next moves.

Research Question

To what extent can a neural network trained on 3×3 Rubik's Cube states up to depth 20 predict strong or optimal next moves toward solving the cube?

More specifically, we want to examine whether curriculum learning, where the model first learns shallow states and then deeper states, helps the network improve its ability to solve or partially solve the cube.

Proposed Method

First, we will represent the Rubik's Cube state in a format that can be used as input to a neural network. For example, each facelet color can be encoded numerically or as a one vector. The output of the model will be one of the possible cube moves, such as U, D, L, R, F, B and their variants.

We will generate training data by creating cube states at different depths from the solved cube. For each state, we will assign a target move that is considered strong or optimal. At

smaller depths, the correct move is easier to know because the cube is close to the solved state. For larger depths, we may use an existing solver or search algorithm to produce good target moves.

The training process will be gradual. We will begin with states of depth 1, then depth 2, and continue increasing the depth. The idea is that the model should first learn simple patterns and then slowly learn more complex situations. This is similar to how humans often learn difficult tasks step by step.

Evaluation

We will evaluate the model in several ways. First, we will measure the prediction accuracy: how often the model predicts the same move as the target move. Second, we will test whether the predicted moves actually help reduce the cube's distance from the solved state. Finally, we will test the model on scrambled cubes and check how many steps it needs to solve or improve the cube state.

We may also compare the model with simpler baselines, such as random move selection or a basic search based method. This will help us understand whether the neural network is really learning useful information.

Expected Outcome

We expect that the model will perform well on states close to the solved cube and gradually improve on deeper states as training progresses. Even if the model does not always find the optimal move, it may still learn to predict moves that bring the cube closer to the solution. This project can show how neural networks, search, and curriculum learning can be combined in a classic AI problem.